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INT305

Probability Review for Machine Learning

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Motivation

Uncertainty arises through:

- Noisy measurements
- Variability between samples 样本间变异性
- Finite size of data sets 数据集有限

Probability provides a consistent framework for the quantification and manipulation of uncertainty.

Sample Space

Sample space Ω is the set of all possible outcomes of an experiment.

Observations $\omega \in \Omega$ are points in the space also called sample outcomes, realizations, or elements.

Events $E \subset \Omega$ are subsets of the sample space.

In this experiment we flip a coin twice:

Sample space All outcomes $\Omega = \{HH, HT, TH, TT\}$

Observation $\omega = HT$ valid sample since $\omega \in \Omega$

Event Both flips same $E = \{HH, TT\}$ valid event since $E \subset \Omega$

} 这个解释
很好

Sample Space

The probability of an event E , $P(E)$, satisfies three axioms:

1: $P(E) \geq 0$ for every E

2: $P(\Omega) = 1$ 把所有可能包含就是1

3: If $E_1, E_2, \dots$ are disjoint then

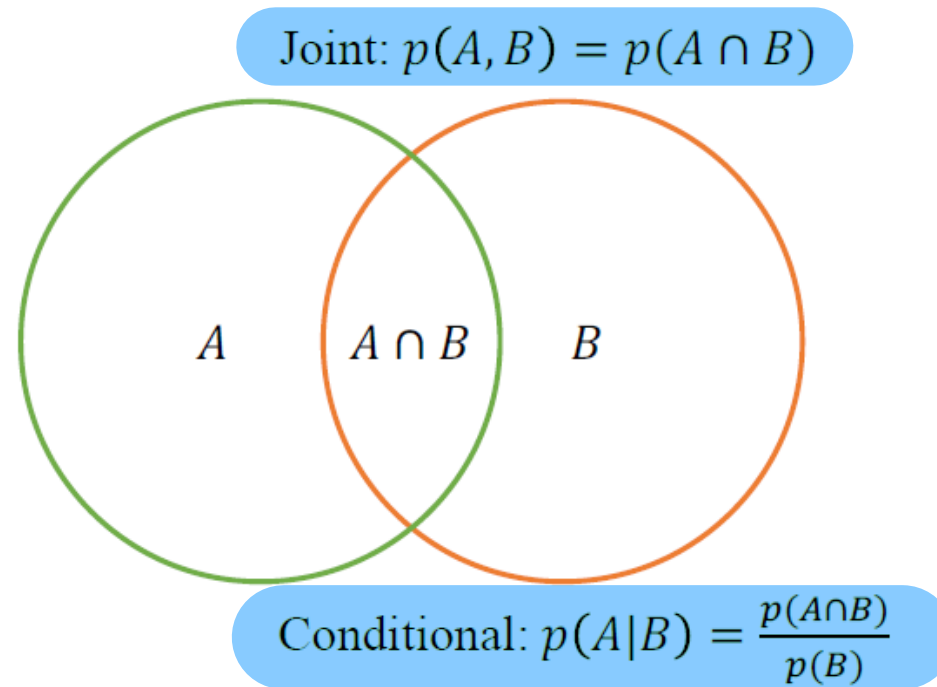
互斥的


$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Joint and Conditional Probabilities

Joint Probability of A and B is denoted $P(A, B)$.

Conditional Probability of A given B is denoted $P(A|B)$.



$$p(A, B) = p(A|B)p(B) = p(B|A)p(A)$$

Conditional Example

Probability of passing the midterm is 60% and probability of passing both the final and the midterm is 45%.

What is the probability of passing the final given the student passed the midterm?

M

$$\begin{aligned}P(F|M) &= P(M, F)/P(M) \\&= 0.45/0.60 \\&= 0.75\end{aligned}$$

$$P(F|M) = \frac{P(F, M)}{P(M)} = \frac{45\%}{60\%} = \frac{3}{4} = 75\%$$

Independence

Events A and B are independent if $P(A, B) = P(A)P(B)$.

- Independent: A : first toss is HEAD; B : second toss is HEAD;

$$P(A, B) = 0.5 * 0.5 = P(A)P(B)$$

- Not Independent: A : first toss is HEAD; B : first toss is HEAD;

$$P(A, B) = 0.5 \neq P(A)P(B)$$

这个例子不好别者

Independence

Events A and B are conditionally independent given C if

$$P(A, B|C) = P(B|C)P(A|C)$$

Consider two coins ²: A regular coin and a coin which always outputs HEAD or always outputs TAIL.

A =The first toss is HEAD; B =The second toss is HEAD; C =The regular coin is used. D =The other coin is used.

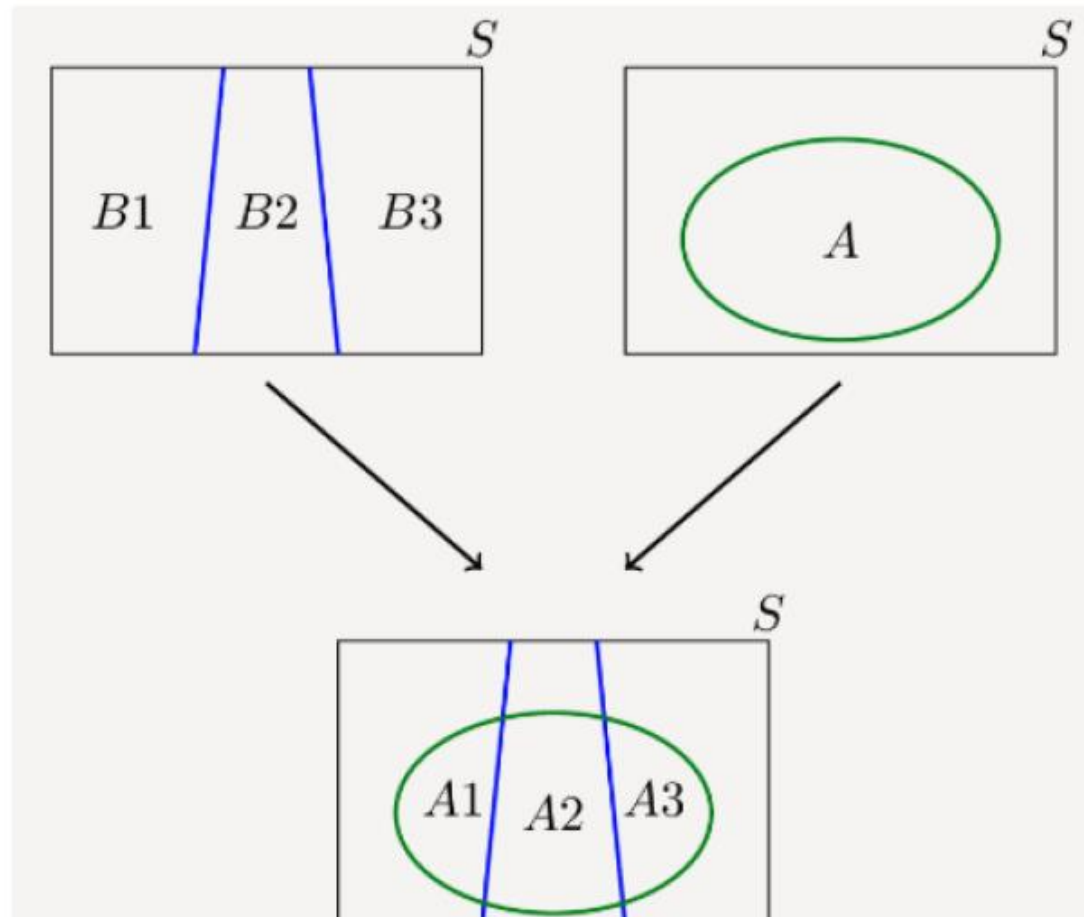
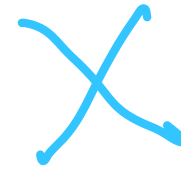
Then A and B are conditionally independent given C , but A and B are NOT conditionally independent given D .

↳ 这里只答条件独立

Marginalization and Law of Total Probability

Law of Total Probability ³

$$P(X) = \sum_Y P(X, Y) = \sum_Y P(X|Y)P(Y)$$



Bayes' Rule

Bayes' Rule:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(\theta|x) = \frac{P(x|\theta)P(\theta)}{P(x)}$$

$$\text{Posterior} = \frac{\text{Likelihood} * \text{Prior}}{\text{Evidence}}$$

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

Bayes' Example

Suppose you have tested positive for a disease. What is the probability you actually have the disease?

$$P(T=0|D=1) = 0.05$$

This depends on the prior probability of the disease:

$$P(T=0|D=0) = 0.9$$

- $P(T = 1|D = 1) = 0.95$ (likelihood)

- $P(T = 1|D = 0) = 0.10$ (likelihood)

- $P(D = 1) = 0.1$ (prior)

So $P(D = 1|T = 1) = ?$
$$\frac{P(T=1, D=1)}{P(T=1)} = \frac{0.95 \times 0.1}{0.9 \times 0.1 + 0.95 \times 0.1} \approx 51.35\%$$

$$\begin{aligned} P(T=1) &= P(D=0, T=1) + P(D=1, T=1) \\ &= P(T=1|D=0)P(D=0) + P(T=1|D=1)P(D=1) \\ &= 0.1 \times 0.9 + 0.95 \times 0.1 \end{aligned}$$

Bayes' Example

Suppose you have tested positive for a disease. What is the probability you actually have the disease?

$$P(T = 1|D = 1) = 0.95 \text{ (true positive)}$$

$$P(T = 1|D = 0) = 0.10 \text{ (false positive)}$$

$$P(D = 1) = 0.1 \text{ (prior)}$$

So $P(D = 1|T = 1) = ?$

Use Bayes' Rule:

$$P(D = 1|T = 1) = \frac{P(T = 1|D = 1)P(D = 1)}{P(T = 1)} = \frac{0.95 * 0.1}{P(T = 1)} = 0.51$$

$$\begin{aligned} P(T = 1) &= P(T = 1|D = 1)P(D = 1) + P(T = 1|D = 0)P(D = 0) \\ &= 0.95 * 0.1 + 0.1 * 0.90 = 0.185 \end{aligned}$$

Random Variable



How do we connect sample spaces and events to data?

A **random variable** is a mapping which assigns a real number $X(\omega)$ to each observed outcome $\omega \in \Omega$

For example, let's flip a coin 10 times. $X(\omega)$ counts the number of Heads we observe in our sequence. If $\omega = HHTHTHHTHT$ then $X(\omega) = 6$.

Discrete and Continuous Random Variables

Discrete Random Variables (骰子只有6个结果)

- Takes countably many values, e.g., number of heads
- Distribution defined by probability mass function (PMF)
- Marginalization: $p(x) = \sum_y p(x, y)$

Continuous Random Variables (手机续航时间: 无限可能)

- Takes uncountably many values, e.g., time to complete task
- Distribution defined by probability density function (PDF)
- Marginalization: $p(x) = \int_y p(x, y) dy$

I.I.D. 独立同分布

Random variables are said to be **independent and identically distributed** (i.i.d.) if they are sampled from the same probability distribution and are mutually independent.

This is a common assumption for observations. For example, coin flips are assumed to be iid.

Probability Distribution Statistics

Mean: First Moment, μ

$$E[x] = \sum_{i=1}^{\infty} x_i p(x_i) \quad (\text{univariate discrete r.v.})$$

$$E[x] = \int_{-\infty}^{\infty} x p(x) dx \quad (\text{univariate continuous r.v.})$$

Variance: Second (central) Moment, σ^2

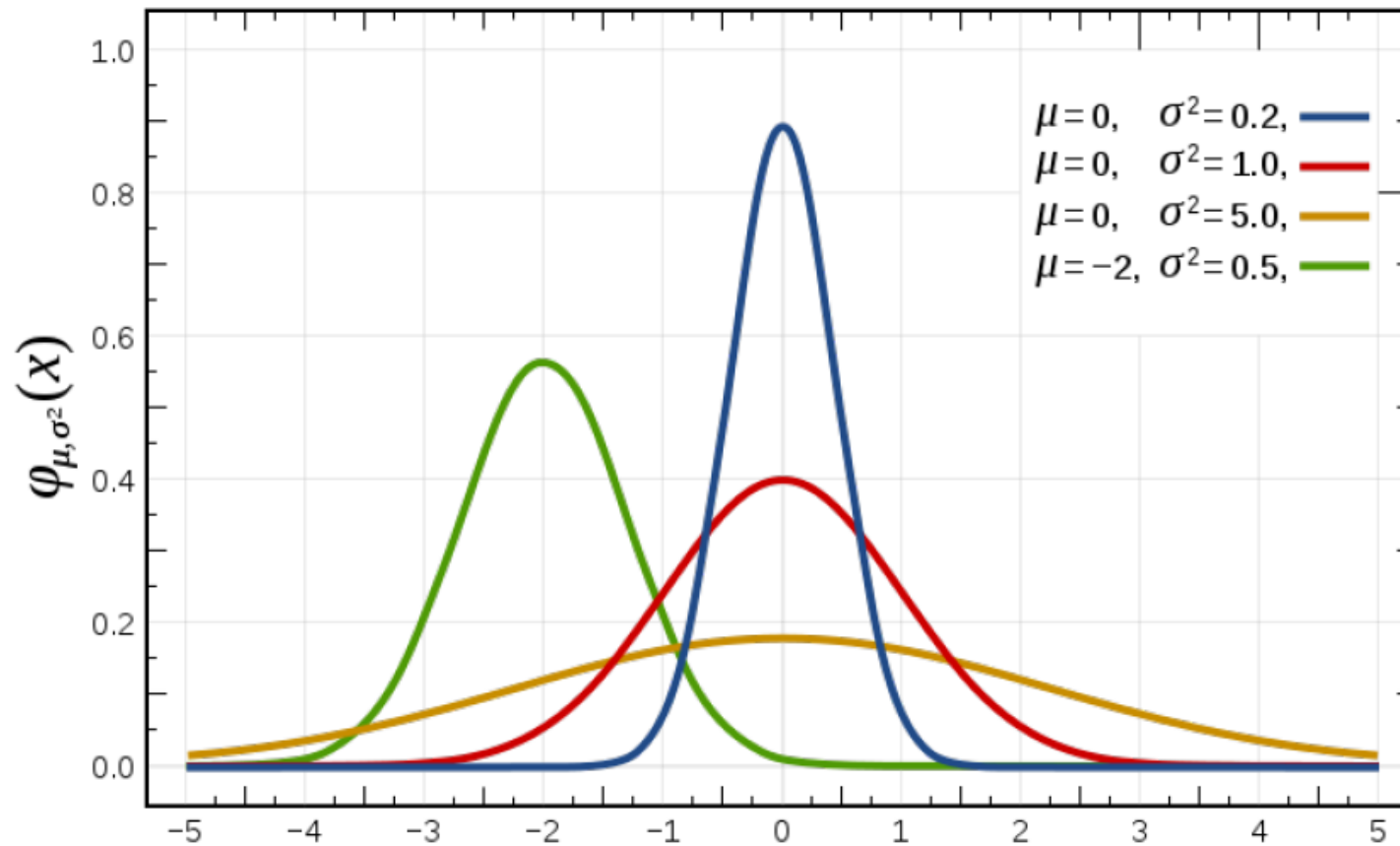
$$\begin{aligned} Var[x] &= \int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx \\ &= E[(x - \mu)^2] \\ &= E[x^2] - E[x]^2 \end{aligned}$$

Univariate Gaussian Distribution

Also known as the **Normal Distribution**, $\mathcal{N}(\mu, \sigma^2)$

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\}$$

单变量的



Multivariate Gaussian Distribution

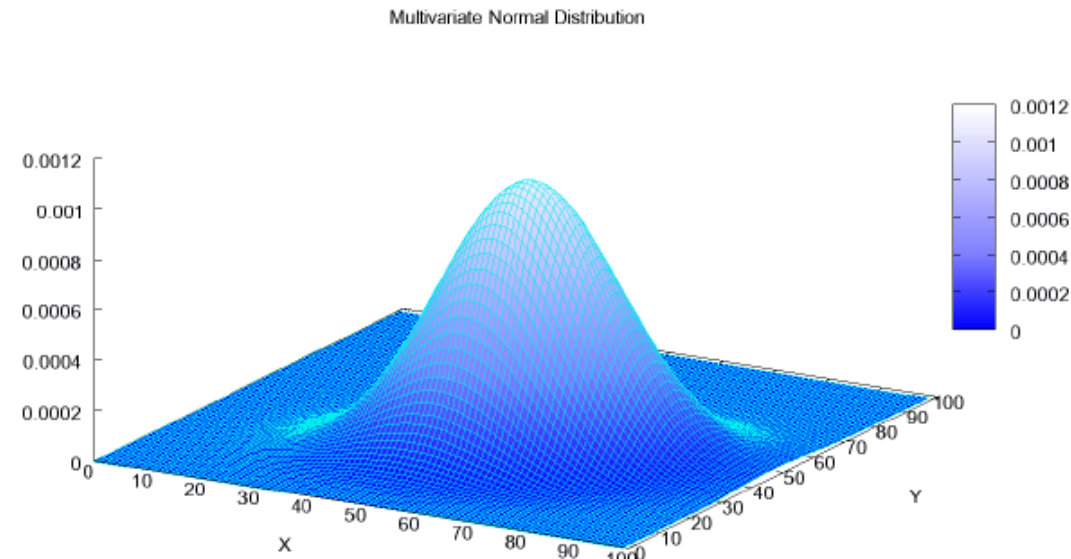
Multidimensional generalization of the Gaussian.

$\mathbf{x}$ is a D -dimensional vector

μ is a D -dimensional mean vector

Σ is a $D \times D$ covariance matrix with determinant $|\Sigma|$

$$\mathcal{N}(\mathbf{x}|\mu, \Sigma) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right\}$$



Covariance Matrix

Recall that $\mathbf{x}$ and μ are D-dimensional vectors

Covariance matrix Σ is a matrix whose (i, j) entry is the covariance

$$\begin{aligned}\Sigma_{ij} &= Cov(\mathbf{x}_i, \mathbf{x}_j) \\ &= E[(\mathbf{x}_i - \mu_i)(\mathbf{x}_j - \mu_j)] \\ &= E[(\mathbf{x}_i \mathbf{x}_j)] - \mu_i \mu_j\end{aligned}$$

so notice that the diagonal entries are the variance of each elements. The covariant matrix has the property that it is symmetric and positive-semidefinite (this is useful for whitening).

Inferring Parameters

We have data X and we assume it is sampled from some distribution.
How do we figure out the parameters that 'best' fit that distribution?
Maximum Likelihood Estimation (MLE)

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta} P(X|\theta) \text{ 找到使 } P(X|\theta) \text{ 最大 } \theta$$

Maximum A posteriori Probability (MAP)

$$\hat{\theta}_{MAP} = \operatorname{argmax}_{\theta} P(\theta|X) \propto P(X|\theta) P(\theta)$$

(引入参数 θ 的先验分布
→ 找到使 posterior $P(\theta|X)$ 最大 θ

MLE for Univariate Gaussian Distribution

We are trying to infer the parameters for a Univariate Gaussian Distribution, mean (μ) and variance (σ^2).

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\}$$

The **likelihood** that our observations $x_1, \dots, x_N$ were generated by a univariate Gaussian with parameters μ and σ^2 is

$$\text{Likelihood} = p(x_1 \dots x_N | \mu, \sigma^2) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x_i - \mu)^2\right\}$$

MLE for Univariate Gaussian Distribution

For MLE we want to maximize this likelihood, which is difficult because it is represented by a product of terms

$$\text{Likelihood} = p(x_1 \dots x_N | \mu, \sigma^2) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x_i - \mu)^2\right\}$$

So we take the log of the likelihood so the product becomes a sum

$$\begin{aligned} \text{Log Likelihood} &= \log p(x_1 \dots x_N | \mu, \sigma^2) \\ &= \sum_{i=1}^N \log \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x_i - \mu)^2\right\} \end{aligned}$$

Since log is monotonically increasing $\max L(\theta) = \max \log L(\theta)$

MLE for Univariate Gaussian Distribution

The log Likelihood simplifies to

$$\begin{aligned}\mathcal{L}(\mu, \sigma) &= \sum_{i=1}^N \log \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x_i - \mu)^2\right\} \\ &= -\frac{1}{2}N \log(2\pi\sigma^2) - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2}\end{aligned}$$

Which we want to maximize. How?

MLE for Univariate Gaussian Distribution

To maximize we take the derivatives, set equal to 0, and solve:

$$\mathcal{L}(\mu, \sigma) = -\frac{1}{2}N \log(2\pi\sigma^2) - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2}$$

Derivative w.r.t. μ , set equal to 0, and solve for $\hat{\mu}$

$$\frac{\partial \mathcal{L}(\mu, \sigma)}{\partial \mu} = 0 \implies \hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$$

Therefore the $\hat{\mu}$ that maximizes the likelihood is the average of the data points.

Derivative w.r.t. σ^2 , set equal to 0, and solve for $\hat{\sigma}^2$

$$\frac{\partial \mathcal{L}(\mu, \sigma)}{\partial \sigma^2} = 0 \implies \hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2$$